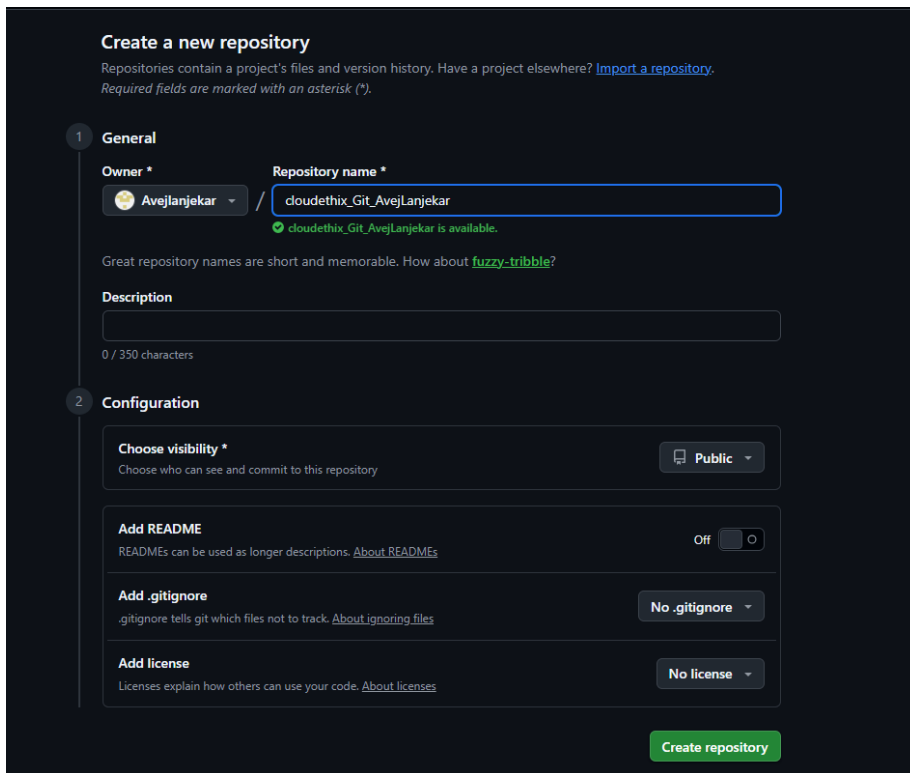


Lab 02

1. Create a remote "repository" with project name cloudeithx_git_yourname. Create one file named demo.txt in the master branch of the remote repository. Once a file is created , clone the repository locally. On the local repository create 3 branches: feature/login and feature/mainline, release. Add login.txt file to feature/login branch and push that branch to remote repository. Create PR and approve the PR and then merge the feature/login branch to the remote master branch.

Step 1 :- create a remote repository with name cloudeithx_Git_AvejLanjekar



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * **Repository name ***

 AvejLanjekar / cloudeithx_Git_AvejLanjekar

✔ cloudeithx_Git_AvejLanjekar is available.

Great repository names are short and memorable. How about [fuzzy-tribble?](#)

Description

0 / 350 characters

2 Configuration

Choose visibility *
Choose who can see and commit to this repository

Public

Add README
READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

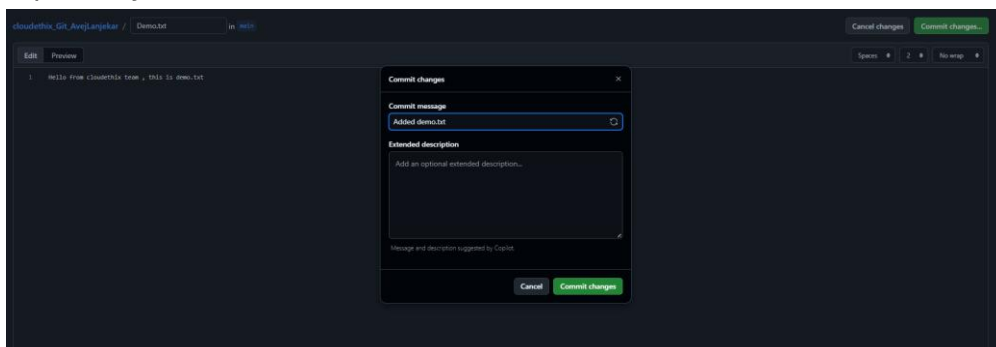
No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Step 2:- Create one file named “demo.txt” in the master branch of the remote repository.



cloudeithx_Git_AvejLanjekar / Demo.txt in main

Cancel changes Commit changes

1 2 No exp

1 Hello from cloudeithx team - this is demo.txt

Commit changes

Commit message
Added demo.txt

Extended description
Add an optional extended description...

Message and description suggested by Copilot

Cancel Commit changes

Step 3:- clone the repository locally.

```
avej_lanjekar@AvejLanjekar:~/git-workshop$ git clone https://github.com/Avejlanjekar/cloudethix_Git_AvejLanjekar.git
Cloning into 'cloudethix_Git_AvejLanjekar'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
avej_lanjekar@AvejLanjekar:~/git-workshop$ ls
cloudethix_Git_AvejLanjekar
avej_lanjekar@AvejLanjekar:~/git-workshop$ cd cloudethix_Git_AvejLanjekar/
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ cat Demo.txt
Hello from cloudethix team , this is demo.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

Step 4:- On the local repository create 3 branches: feature/login and feature/mainline, release.

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch
* main
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch feature/login
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch feature/mainline
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch release
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

Step 5:- Add login.txt file to feature/login branch and push that branch to remote repository.

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git checkout feature/login
Switched to branch 'feature/login'
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ echo "Implementing Login feature" > login.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt  login.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git status
On branch feature/login
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    login.txt

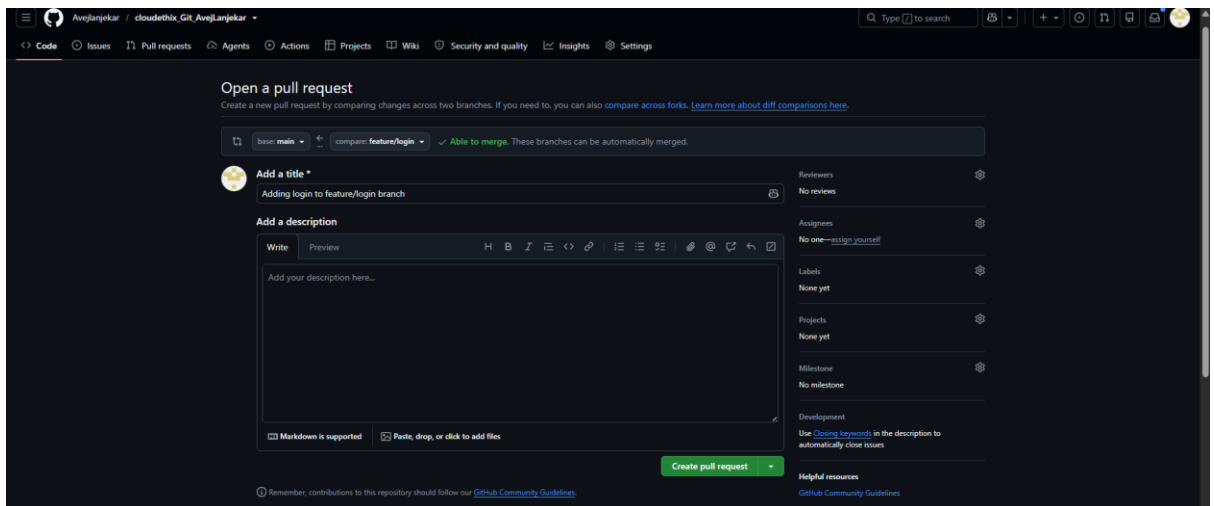
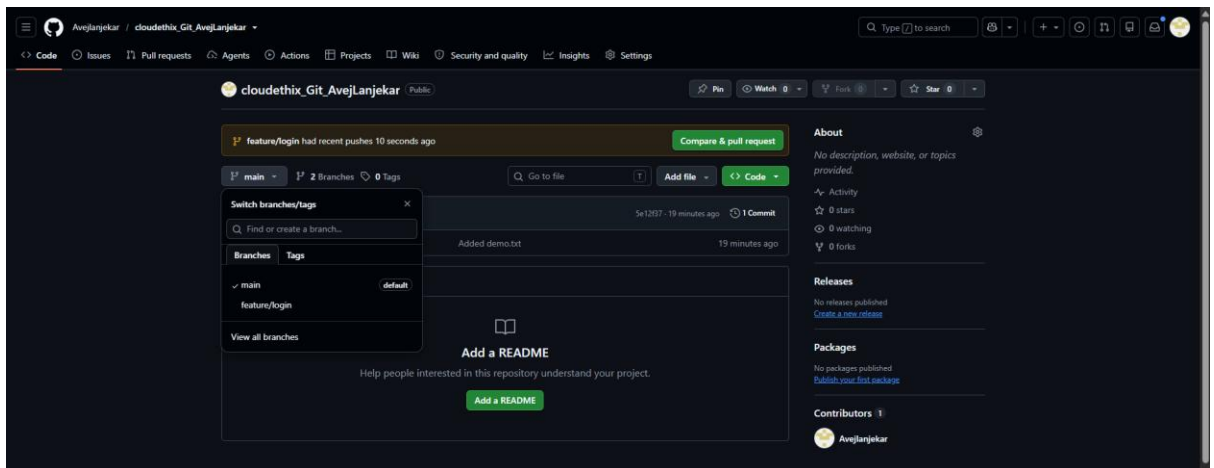
nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git add login.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git commit -m "Adding login to feature/login branch"
[feature/login 89babe5] Adding login to feature/login branch
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git status
On branch feature/login
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

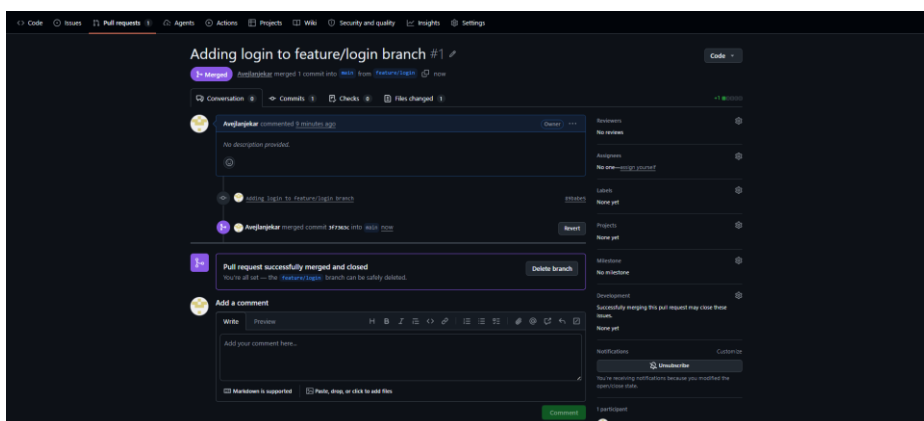
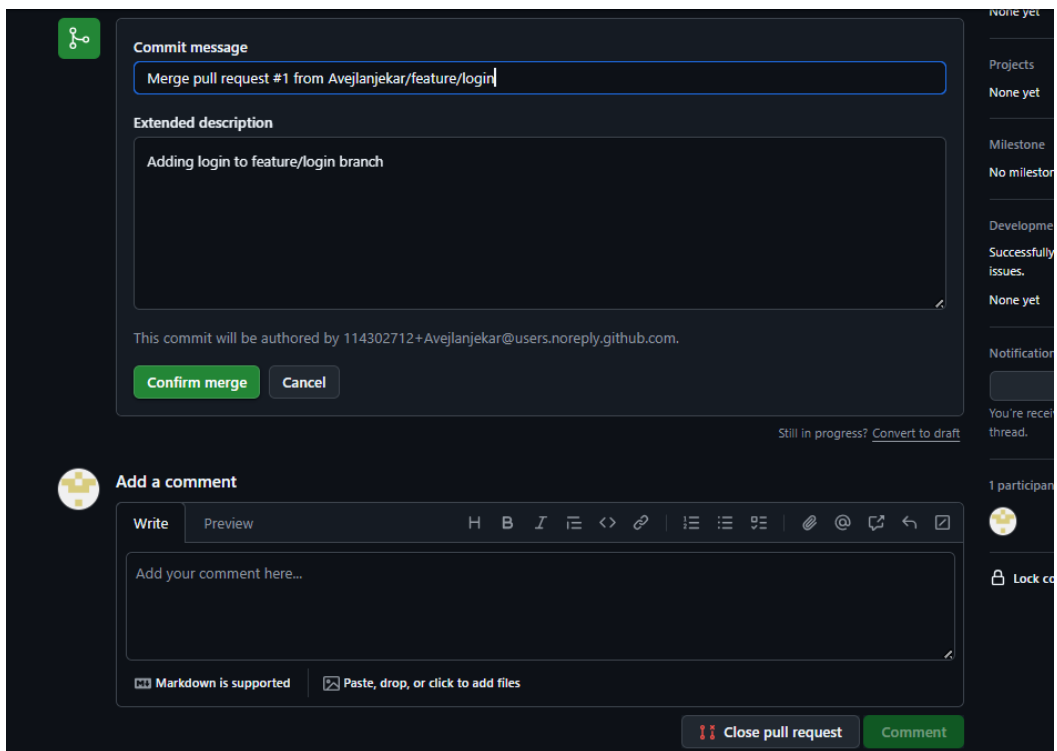
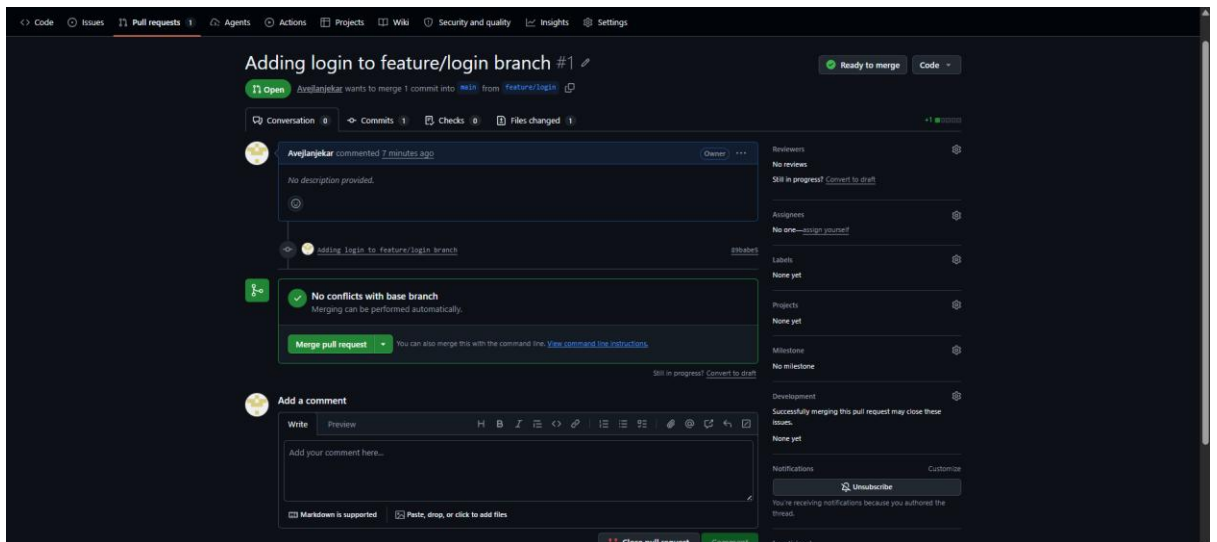
```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch
* feature/login
  feature/mainline
  main
  release
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote -v
origin  https://github.com/Avejlanjekar/cloudethix_Git_AvejLanjekar.git (fetch)
origin  https://github.com/Avejlanjekar/cloudethix_Git_AvejLanjekar.git (push)
```

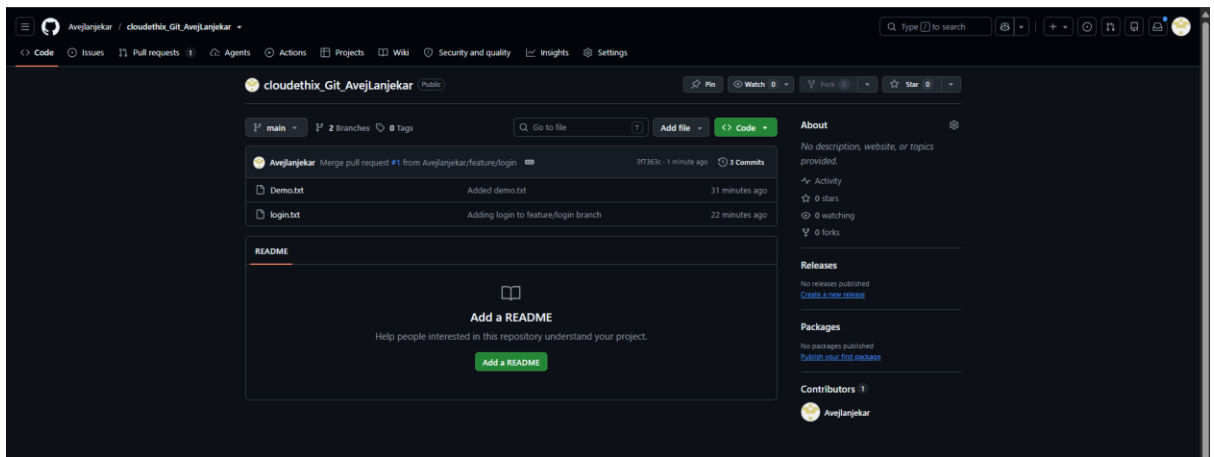
```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote remove origin
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote -v
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote add origin git@github.com:Avejlanjekar/cloudethix_Git_AvejLanjekar.git
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote -v
origin  git@github.com:Avejlanjekar/cloudethix_Git_AvejLanjekar.git (fetch)
origin  git@github.com:Avejlanjekar/cloudethix_Git_AvejLanjekar.git (push)
```

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git push origin feature/login
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 322 bytes | 80.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/login' on GitHub by visiting:
remote:   https://github.com/Avejlanjekar/cloudethix_Git_AvejLanjekar/pull/new/feature/login
remote:
To github.com:Avejlanjekar/cloudethix_Git_AvejLanjekar.git
 * [new branch]      feature/login -> feature/login
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

Step 6 :- Create PR and approve the PR and then merge the feature/login branch to the remote master branch.







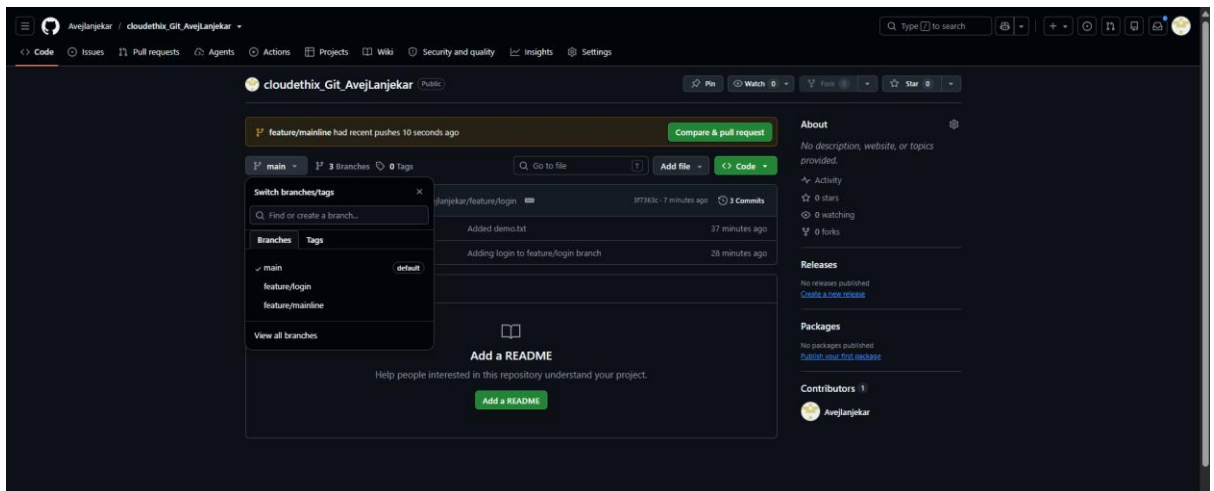
2. Add files to the feature/mainline branch to local repo and push that branch to remote. Then add some files to the feature/mainline remote branch from the UI. Then fetch the latest changes on local feature/mainline with Fetch and Pull command.

Step 1:- Add files to the feature/mainline branch to local repo and push that branch to remote

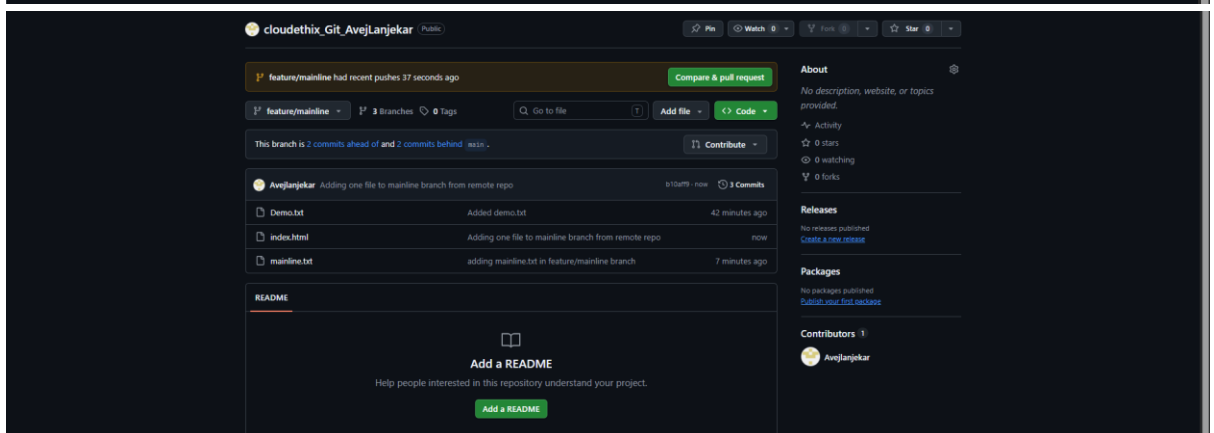
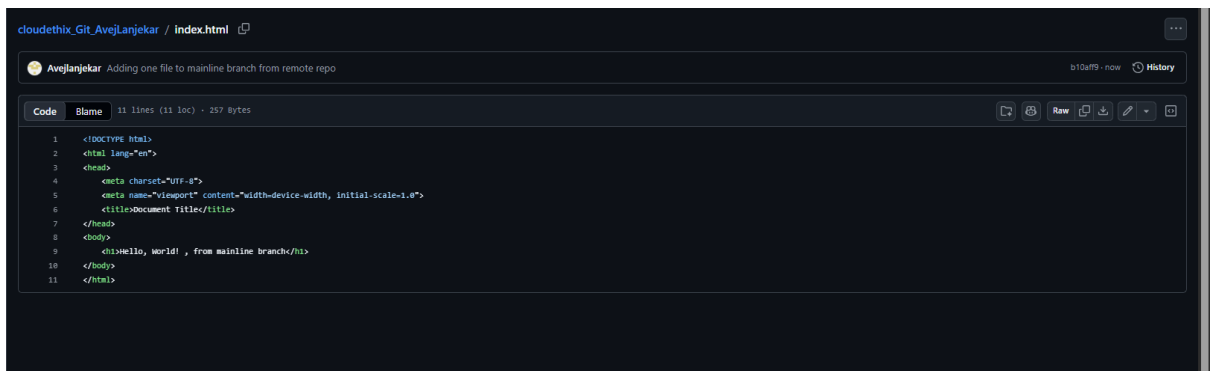
```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch
* feature/login
  feature/mainline
  main
  release
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git checkout feature/mainline
Switched to branch 'feature/mainline'
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ echo "Working on mainline branch" > mainline.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git status
On branch feature/mainline
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    mainline.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git add mainline.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git commit -m "adding mainline.txt in feature/mainline branch"
[feature/mainline 6555b6f] adding mainline.txt in feature/mainline branch
 1 file changed, 1 insertion(+)
 create mode 100644 mainline.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git status
On branch feature/mainline
nothing to commit, working tree clean
```

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git remote -v
\origin git@github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar.git (fetch)
origin git@github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar.git (push)
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git push origin feature/mainline
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 333 bytes | 111.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/mainline' on GitHub by visiting:
remote:   https://github.com/AvejLanjekar/cloudethix_Git_AvejLanjekar/pull/new/feature/mainline
remote:
To github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar.git
 * [new branch]      feature/mainline -> feature/mainline
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```



Step 2:- Add some files to the feature/mainline remote branch from the UI

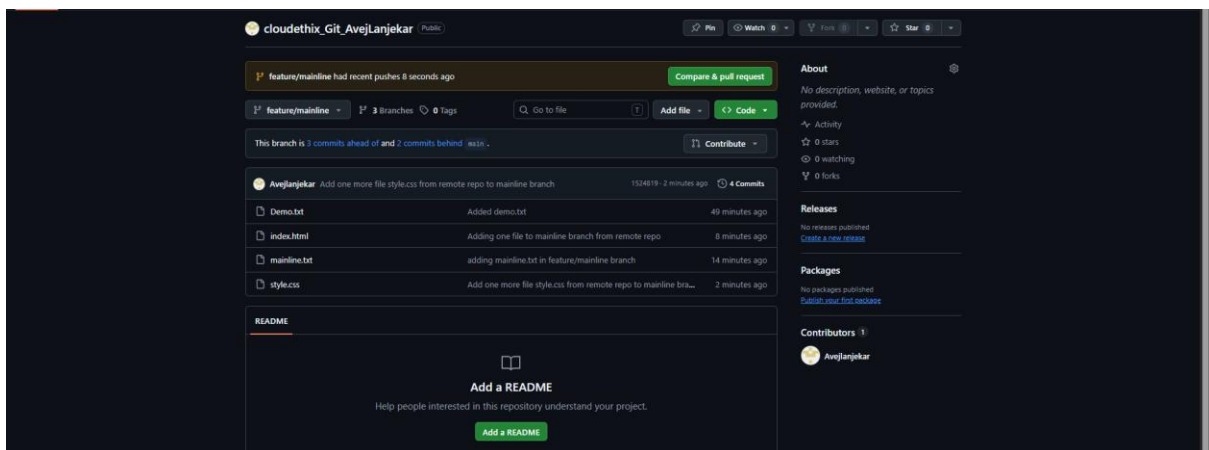


Step 3:- fetch the latest changes on local feature/mainline with Fetch and Pull command.

Using git pull

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch
feature/login
* feature/mainline
main
release
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git pull origin feature/mainline
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1.15 KiB | 130.00 KiB/s, done.
From github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar
* branch                feature/mainline -> FETCH_HEAD
6555b6f..b10aff9 feature/mainline -> origin/feature/mainline
Updating 6555b6f..b10aff9
Fast-forward
 index.html | 11 ++++++++
 1 file changed, 11 insertions(+)
 create mode 100644 index.html
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git status
On branch feature/mainline
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt index.html mainline.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document Title</title>
</head>
<body>
  <h1>Hello, World! , from mainline branch</h1>
</body>
</html>
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

Adding files from remote repo to feature/mainline to download latest changes on local repo using git fetch



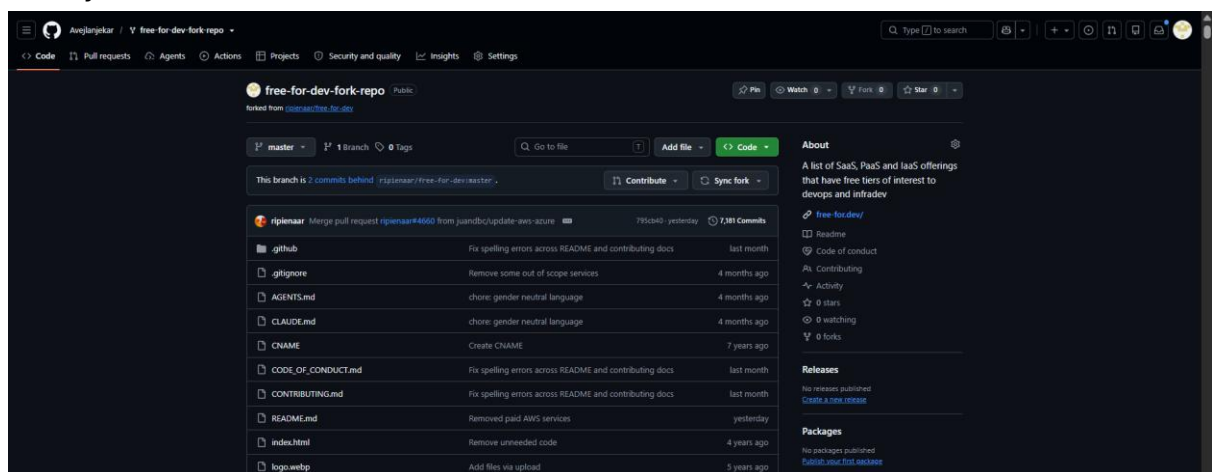
Added style.css from remote repo

Using git fetch to download changes

```
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git fetch origin feature/mainline
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1003 bytes | 250.00 KiB/s, done.
From github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar
* branch          feature/mainline -> FETCH_HEAD
b10aff9..1524819 feature/mainline -> origin/feature/mainline
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt  index.html  mainline.txt
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git branch
* feature/login
* feature/mainline
main
release
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ git pull origin feature/mainline
From github.com:AvejLanjekar/cloudethix_Git_AvejLanjekar
* branch          feature/mainline -> FETCH_HEAD
Updating b10aff9..1524819
Fast-forward
 style.css | 5 +++++
1 file changed, 5 insertions(+)
create mode 100644 style.css
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ ls
Demo.txt  index.html  mainline.txt  style.css
avej_lanjekar@AvejLanjekar:~/git-workshop/cloudethix_Git_AvejLanjekar$ |
```

3. Fork below public repository to your account. Then clone the repository locally. Create 2 branches cloud_dev01. Then add cloud_dev02 to that repository locally and hello.txt , moto.txt and world.txt to one branch and add another 3 files to another branch. Then push cloud_dev02 branch to the Forked repository on the remote. [GitHub - ripienaar/free-for-dev: A list of SaaS, PaaS and IaaS offerings that have free tiers of int...](#)

Step 1:- Fork below public repository to your account & clone the repository locally.



```
avej_lanjekar@AvejLanjekar:~$ cd git_fork_workshop/
avej_lanjekar@AvejLanjekar:~/git_fork_workshop$ git clone https://github.com/AvejLanjekar/free-for-dev-fork-repo.git
Cloning into 'free-for-dev-fork-repo'...
remote: Enumerating objects: 16923, done.
remote: Counting objects: 100% (43/43), done.
remote: Compressing objects: 100% (22/22), done.
remote: Total 16923 (delta 34), reused 21 (delta 21), pack-reused 16880 (from 3)
Receiving objects: 100% (16923/16923), 10.11 MiB | 3.83 MiB/s, done.
Resolving deltas: 100% (9007/9007), done.
avej_lanjekar@AvejLanjekar:~/git_fork_workshop$ ls
free-for-dev-fork-repo
avej_lanjekar@AvejLanjekar:~/git_fork_workshop$ cd free-for-dev-fork-repo/
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ ls
AGENTS.md  CLAUDE.md  CNAME  CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  index.html  logo.webp
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ |
```


Step 2:- Create 2 branches cloud_dev01 and cloud_dev02

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch
* master
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch cloud_dev01
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch cloud_dev02
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch
cloud_dev01
cloud_dev02
* master
```

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git checkout cloud_dev01
Switched to branch 'cloud_dev01'
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch
* cloud_dev01
cloud_dev02
master
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ touch hello.txt moto.txt world.txt
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev01
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.txt
    moto.txt
    world.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git add hello.txt moto.txt world.txt
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev01
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   hello.txt
    new file:   moto.txt
    new file:   world.txt

avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git commit -m "adding files to cloud_dev01 branch"
[cloud_dev01 de928898] adding files to cloud_dev01 branch
3 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.txt
create mode 100644 moto.txt
create mode 100644 world.txt
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev01
nothing to commit, working tree clean
```

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git checkout cloud_dev02
Switched to branch 'cloud_dev02'
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ ls
AGENTS.md  CNAME          CONTRIBUTING.md  index.html
CLAUDE.md  CODE_OF_CONDUCT.md  README.md       Logo.webp
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ touch file1.txt file2.txt file3.txt
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev02
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    file1.txt
    file2.txt
    file3.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git add .
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git commit -m "adding files to cloud_dev02 branch"
[cloud_dev02 b8b3dc5d] adding files to cloud_dev02 branch
3 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file1.txt
create mode 100644 file2.txt
create mode 100644 file3.txt
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev02
nothing to commit, working tree clean
```

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch cloud_dev02
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git remote -v
origin  https://github.com/Avejlanjekar/free-for-dev-fork-repo.git (fetch)
origin  https://github.com/Avejlanjekar/free-for-dev-fork-repo.git (push)
```

```

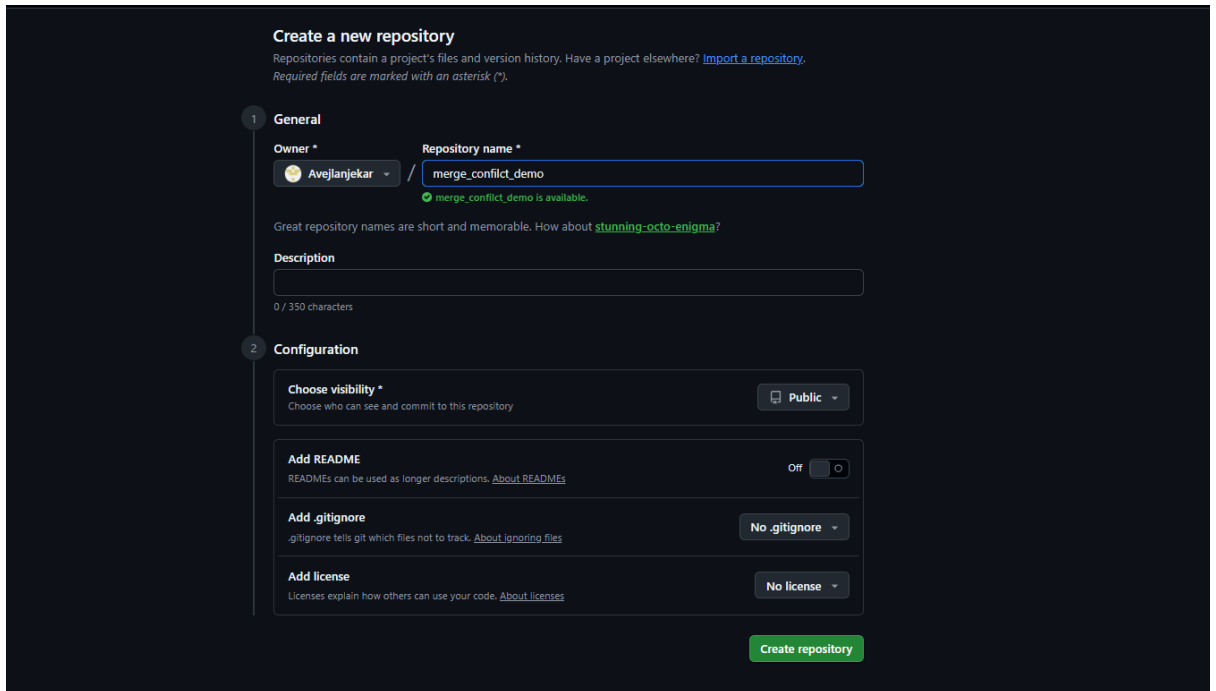
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git remote remove origin
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git remote add origin git@github.com:Avejlanjekar/free-for-dev-fork-repo.git
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git remote -v
origin  git@github.com:Avejlanjekar/free-for-dev-fork-repo.git (fetch)
origin  git@github.com:Avejlanjekar/free-for-dev-fork-repo.git (push)
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git push origin cloud_dev02
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 307 bytes | 153.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'cloud_dev02' on GitHub by visiting:
remote:   https://github.com/Avejlanjekar/free-for-dev-fork-repo/pull/new/cloud_dev02
remote:
To github.com:Avejlanjekar/free-for-dev-fork-repo.git
 * [new branch]      cloud_dev02 -> cloud_dev02
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$

```

The screenshot shows the GitHub repository page for 'free-for-dev-fork-repo'. The repository is public and was forked from 'diancar/free-for-dev'. The main branch is 'master', and there are 2 branches and 0 tags. The repository has 7,181 commits and 0 forks. The repository is currently on the 'cloud_dev02' branch, which was pushed 5 minutes ago. The repository has a README, a Code of Conduct, and a CONTRIBUTING file. The repository is currently on the 'cloud_dev02' branch, which was pushed 5 minutes ago. The repository has a README, a Code of Conduct, and a CONTRIBUTING file. The repository is currently on the 'cloud_dev02' branch, which was pushed 5 minutes ago. The repository has a README, a Code of Conduct, and a CONTRIBUTING file.

4. Create a remote repository merge_conflict_demo. Add some files in the remote master branch. Clone the repository locally. Create 2 branches locally named demo/merge and demo/conflicts. Add a merge.txt file in both branches with some data like "Hello CloudEthiX...." And "Fello CloudEthiX...." respectively and commit. Then merge demo/conflicts branch to demo/merge branch and solve the merge conflicts.

Step 1:- Create a remote repository merge_conflict_demo. Add some files in the remote master branch. Clone the repository locally.



The screenshot shows the 'Create a new repository' page on GitHub. The 'General' tab is selected, showing the repository name 'merge_conflict_demo' and a description field. The 'Configuration' tab is also visible, showing options for visibility (Public), adding a README, .gitignore, and a license. A green 'Create repository' button is at the bottom right.

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * Avejlanjekar / Repository name *
merge_conflict_demo is available.

Great repository names are short and memorable. How about [stunning-octo-enigma?](#)

Description
0 / 350 characters

2 Configuration

Choose visibility * Public
Choose who can see and commit to this repository

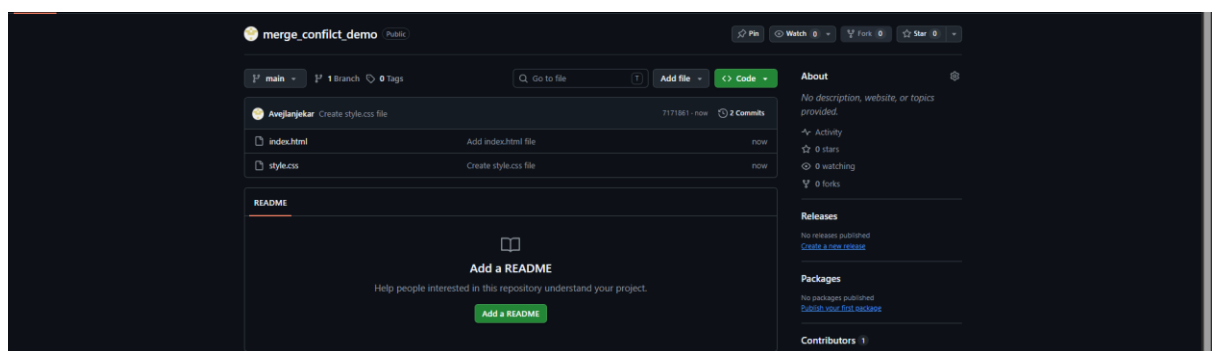
Add README Off
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license
Licenses explain how others can use your code. [About licenses](#)

[Create repository](#)

Added some files in remote master branch



Clone the repo locally:-

```
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo$ git clone https://github.com/AvejLanjekar/merge_conflict_demo.git
Cloning into 'merge_conflict_demo'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo$ ls
merge_conflict_demo
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo$ cd merge_conflict_demo/
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ ls
index.html  style.css
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ |
```

Step 2:- Create 2 branches locally named demo/merge and demo/conflicts.

```
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git branch
* main
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git branch demo/merge
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git branch demo/conflicts
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git branch
demo/conflicts
demo/merge
* main
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ |
```

Step 3:- Add a merge.txt file in both branches with some data like "Hello CloudEthiX...." And "Fello CloudEthiX...." respectively and commit.

```
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git checkout merge
error: pathspec 'merge' did not match any file(s) known to git
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git checkout demo/merge
Switched to branch 'demo/merge'
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ ls
index.html  style.css
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ echo "Hello cloudethix" > merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git status
On branch demo/merge
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    merge.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git add merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git commit -m "adding merge.txt in demo/merge branch"
[demo/merge b458b6b] adding merge.txt in demo/merge branch
1 file changed, 1 insertion(+)
create mode 100644 merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git status
On branch demo/merge
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$
```

```
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git checkout demo/conflicts
Switched to branch 'demo/conflicts'
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ ls
index.html  style.css
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ echo "Fello cloudethix" > merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git add merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git commit -m "adding merge.txt in demo/conflicts branch"
[demo/conflicts 21798f8] adding merge.txt in demo/conflicts branch
1 file changed, 1 insertion(+)
create mode 100644 merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_conflict_demo$ git status
On branch demo/conflicts
nothing to commit, working tree clean
```

Step 4:- merge demo/conflicts branch to demo/merge branch and solve the merge conflicts.

```
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git checkout demo/merge
Switched to branch 'demo/merge'
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ ls
index.html  merge.txt  style.css
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ cat merge.txt
Hello cloudeithix
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git merge demo/conflicts
Auto-merging merge.txt
CONFLICT (add/add): Merge conflict in merge.txt
Automatic merge failed; fix conflicts and then commit the result.
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ cat merge.txt
<<<<<< HEAD
Hello cloudeithix
=====
Fellow cloudeithix
>>>>>> demo/conflicts
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ vim merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ cat merge.txt
Hello cloudeithix Fellow
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git status
On branch demo/merge
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both added:   merge.txt

no changes added to commit (use "git add" and/or "git commit -a")
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git add merge.txt
avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git status
On branch demo/merge
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   merge.txt

avej_lanjekar@AvejLanjekar:~/merge_conflict_demo/merge_confilct_demo$ git commit -m "solved merge conflicts of file merge.txt"
```

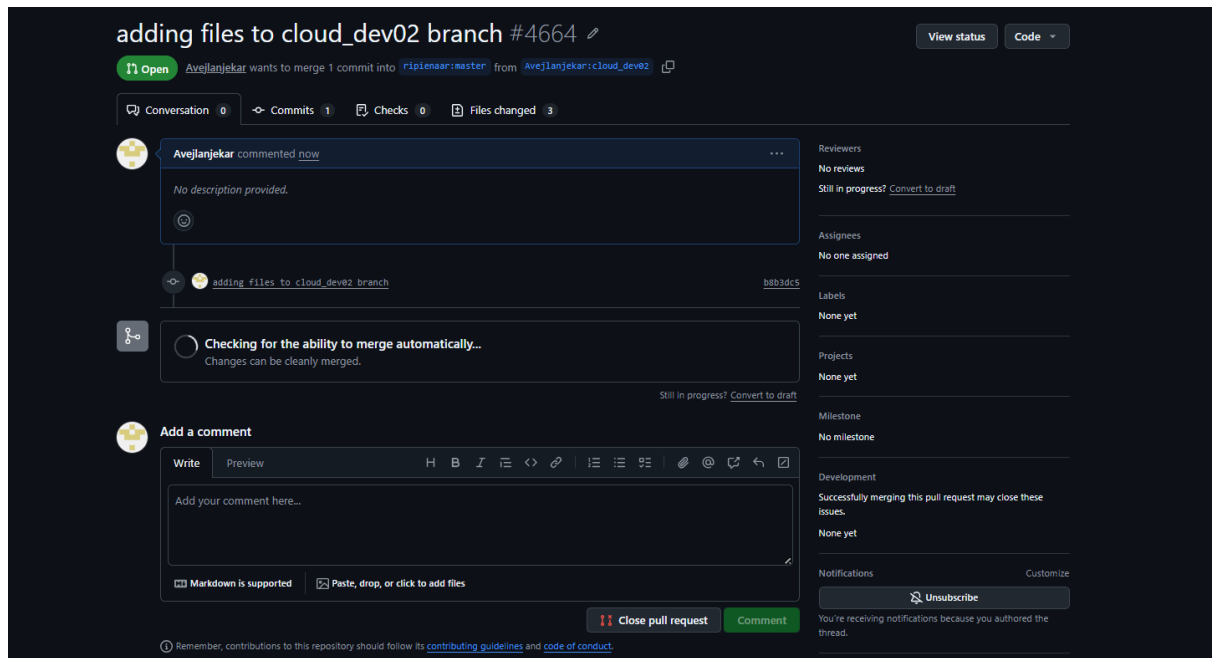
5. Create PR to merge the cloud_dev02 branch to Forked repository. Approve the PR and merge cloud_dev02 branch to master branch remotely.

The screenshot shows the GitHub repository page for 'free-for-dev-fork-repo', which is a fork of 'ripienaar/free-for-dev'. The repository is public and has 0 stars and 0 forks. The main branch is 'master', and there are 2 branches and 0 tags. A notification bar at the top indicates that 'cloud_dev02' had recent pushes 44 minutes ago, with a 'Compare & pull request' button. Below the notification, there is a table of files and their commit history:

File	Commit Message	Time
.github	Fix spelling errors across README and contributing docs	last month
.gitignore	Remove some out of scope services	4 months ago
AGENTS.md	chore: gender neutral language	4 months ago
CLAUDE.md	chore: gender neutral language	4 months ago
CNAME	Create CNAME	7 years ago
CODE_OF_CONDUCT.md	Fix spelling errors across README and contributing docs	last month
CONTRIBUTING.md	Fix spelling errors across README and contributing docs	last month
README.md	Removed paid AWS services	yesterday
index.html	Remove unneeded code	4 years ago
logo.webp	Add files via upload	5 years ago

On the right side, there is an 'About' section with a description of the repository and links to the README, Code of conduct, and Contributing guidelines. Below this, there are sections for 'Releases', 'Packages', and 'Contributors', all of which show 'No releases published' or 'No contributors'.

The screenshot shows the 'Comparing changes' page for a pull request. The 'base repository' is 'ripienaar/free-for-dev' and the 'base' branch is 'master'. The 'head repository' is 'Avejanjekar/free-for-dev-fork-...' and the 'compare' branch is 'cloud_dev02'. A green checkmark indicates that the pull request is 'Able to merge' and that the branches can be automatically merged. Below this, there is a section for 'Add a title' with the text 'adding files to cloud_dev02 branch'. Underneath, there is a section for 'Add a description' with a text area for the description. At the bottom, there is a checkbox for 'Allow edits by maintainers' which is checked, and a 'Create pull request' button. A tooltip on the right side of the page says: 'It looks like this is your first time opening a pull request in this project! Be sure to review the contributing guidelines and code of conduct.'



6. **Fetch latest changes from master branch of remote forked repository to local master branch. Then merge forked remote master branch with local master branch. Check If all the latest files are there. Then checkout to local cloud_dev03 branch locally. Then Pull latest changes from the remote master branch.**

Step 1:- Fetch latest changes from master branch of remote forked repository to local master branch. Then merge forked remote master branch with local master branch

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git fetch origin
From github.com:Avejlanjekar/free-for-dev-fork-repo
* [new branch]      master    -> origin/master
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ ls
AGENTS.md  CLAUDE.md  CNAME      CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  index.html  logo.webp
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git status
On branch master
nothing to commit, working tree clean
```

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git merge origin/master
Already up to date.
```

Step 2:- Check If all the latest files are there. Then checkout to local cloud_dev03 branch locally. Then Pull latest changes from the remote master branch.

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git checkout cloud_dev02
Switched to branch 'cloud_dev02'
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ ls
AGENTS.md  CLAUDE.md  CNAME  CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  file1.txt  file2.txt  file3.txt  index.html  logo.webp
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git branch
  cloud_dev01
* cloud_dev02
  master
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git pull origin master
From github.com:AvejLanjekar/free-for-dev-fork-repo
* branch      master       -> FETCH_HEAD
Already up to date.
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ ls
AGENTS.md  CLAUDE.md  CNAME  CODE_OF_CONDUCT.md  CONTRIBUTING.md  README.md  file1.txt  file2.txt  file3.txt  index.html  logo.webp
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ |
```

7. Checkout to the cloud_dev01 branch locally and rebase the remote master branch of the forked repository.

```
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git checkout cloud_dev01
Switched to branch 'cloud_dev01'
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git rebase origin/master
Current branch cloud_dev01 is up to date.
avej_lanjekar@AvejLanjekar:~/git_fork_workshop/free-for-dev-fork-repo$ git log
commit de9288982d848bc172e71c8a7dc74c8fd79f353 (HEAD -> cloud_dev01)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date:   Wed Jul 29 05:05:26 2026 +0000

    adding files to cloud_dev01 branch
```

8. Create one repository locally. Commit any 3 files to the local repository. And the revert back the commits.

```
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git_revert_workshop/.git/
```



```

avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ echo "hello cloudeith team 1" > file1.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ echo "hello cloudeith team 2" > file2.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ echo "hello cloudeith team 3" > file3.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ ls
file1.txt file2.txt file3.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file1.txt
        file2.txt
        file3.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git add .
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git commit "Initially Adding 3 files"
error: pathspec 'Initially Adding 3 files' did not match any file(s) known to git
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git commit -m "Initially Adding 3 files"
[master (root-commit) 4208523] Initially Adding 3 files
3 files changed, 3 insertions(+)
create mode 100644 file1.txt
create mode 100644 file2.txt
create mode 100644 file3.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git status
On branch master
nothing to commit, working tree clean

```

```

avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git log
commit 4208523d6158be4d87210ec00f0a01c076937d49 (HEAD -> master)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date:   Wed Jul 29 06:15:38 2026 +0000

    Initially Adding 3 files
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git log --oneline
4208523 (HEAD -> master) Initially Adding 3 files
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git revert 4208523
[master a1cf501] Revert "Initially Adding 3 files"
3 files changed, 3 deletions(-)
delete mode 100644 file1.txt
delete mode 100644 file2.txt
delete mode 100644 file3.txt
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ ls
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ git log --oneline
a1cf501 (HEAD -> master) Revert "Initially Adding 3 files"
4208523 Initially Adding 3 files
avej_lanjekar@AvejLanjekar:~/git_revert_workshop$ |

```

9. Create one local repository. commit some files to the local master branch. Checkout to master_02 branch. Create 2 more files. Add those to the staging area. Create stash area and those files. Then list and delete the stash.

Step 1:- Create one local repository. commit some files to the local master branch.

```
avej_lanjekar@AvejLanjekar:~$ mkdir git_stash_workshop
avej_lanjekar@AvejLanjekar:~$ cd git_stash_workshop/
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ ls
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:     git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:     git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git_stash_workshop/.git/
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ echo "Hello this is Avej Lanjekar" > avej.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ echo "Hello this is cloud" > cloud.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git add .
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git commit -m "Initial commit on master branch adding 2 files"
[master (root-commit) edd8af0] Initial commit on master branch adding 2 files
 2 files changed, 2 insertions(+)
 create mode 100644 avej.txt
 create mode 100644 cloud.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git status
On branch master
nothing to commit, working tree clean
```

Step 2:- Checkout to master_02 branch. Create 2 more files. Add those to the staging area. Create stash area and those files. Then list and delete the stash.

```

avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git branch
* master
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git branch master_02
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git branch
* master
  master_02
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git checkout master_02
Switched to branch 'master_02'
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ ls
avej.txt  cloud.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ echo "recently completed BE" > grad.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ echo "woeking as a trainee" > intern.txt
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git status
On branch master_02
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    grad.txt
    intern.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git add .
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash
Saved working directory and index state WIP on master_02: edd8af0 Initial commit on master branch adding 2 files
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash show
grad.txt | 1 +
intern.txt | 1 +
2 files changed, 2 insertions(+)
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash list
stash@{0}: WIP on master_02: edd8af0 Initial commit on master branch adding 2 files
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash show stash@{0}
grad.txt | 1 +
intern.txt | 1 +
2 files changed, 2 insertions(+)
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash pop stash@{0}
On branch master_02
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   grad.txt
    new file:   intern.txt

Dropped stash@{0} (dd930f79dd0a462a0ef9b69c54899f451bb4646f)

avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash pop stash@{0}
On branch master_02
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   grad.txt
    new file:   intern.txt

Dropped stash@{0} (dd930f79dd0a462a0ef9b69c54899f451bb4646f)
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash show stash@{0}
error: stash@{0} is not a valid reference
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash list
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ git stash show
No stash entries found.
avej_lanjekar@AvejLanjekar:~/git_stash_workshop$ |

```

10. Create a local repository, initialise it and create 3 branches: develop, hotfix and release. Checkout to hotfix branch and commit cloudethix.txt and once committed amend the commit message.

Step 1:- Create a local repository, initialise it and create 3 branches: develop, hotfix and release.

```
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/avej_lanjekar/git_ammend_workshop/.git/
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$
```

```
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git branch
* master
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git branch develop
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git branch hotfix
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git branch release
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$
```

Step 2:- Checkout to hotfix branch and commit cloudethix.txt and once committed amend the commit message.

```
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git checkout hotfix
Switched to branch 'hotfix'
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ ls
file1.txt
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ echo "cloudEthix , located in baner pune" > cloudethix.txt
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ ls
cloudethix.txt  file1.txt
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git status
On branch hotfix
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        cloudethix.txt

nothing added to commit but untracked files present (use "git add" to track)
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git add cloudethix.txt
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git commit "added cloudethix.txt in hotfix branch"
error: pathspec 'added cloudethix.txt in hotfix branch' did not match any file(s) known to git
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git commit -m "added cloudethix.txt in hotfix branch"
[hotfix 213c54a] added cloudethix.txt in hotfix branch
 1 file changed, 1 insertion(+)
 create mode 100644 cloudethix.txt
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git status
On branch hotfix
nothing to commit, working tree clean
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git commit --amend -m "Adding CloudEthix text file in hotfix branch"
[hotfix 18eb1c2] Adding CloudEthix text file in hotfix branch
 Date: Wed Jul 29 06:41:52 2026 +0000
 1 file changed, 1 insertion(+)
 create mode 100644 cloudethix.txt
```

```
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ git log
commit 18eb1c200ebe5da735ecf66cf9721067516ecd17 (HEAD -> hotfix)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date:   Wed Jul 29 06:41:52 2026 +0000

    Adding CloudEthix text file in hotfix branch

commit e89ea60a520e177c5ff6c3e99a06b3c96351be74 (release, master, develop)
Author: Avej K Lanjekar <avej.lanjekar11@gmail.com>
Date:   Wed Jul 29 06:39:02 2026 +0000

    initial commit
avej_lanjekar@AvejLanjekar:~/git_ammend_workshop$ |
```